

Estimation on Subset of Regression Coefficient

Q1) Prove Theorem 1

In the multiple Linear Regression Model, we decompose the design matrix as $X = [X_1, X_2]$. Define a $n \times p_2$ matrix $X_{2,\perp}$ by:

$$X_{2,\perp} = X_2 - \Pi_{X_1} X_2 = (I_n - \Pi_{X_1}) X_2$$

Π_{X_1} is an orthogonal projection onto the column space of X_1

(a) $C(X_{2,\perp})$ is orthogonal to $C(X_1)$

(b) $C(X) = C(X_1) + C(X_{2,\perp})$

Let $X_1 \in \mathbb{R}^{n \times p_1}$ and $X_2 \in \mathbb{R}^{n \times p_2}$ ($p_1 + p_2 = p$)

Let $a \in \mathbb{R}^n$ and $b \in \mathbb{R}^n$

$$\Pi_{X_1} a = X_1 (X_1^T X_1)^{-1} X_1^T a \quad \text{and} \quad \Pi_{X_{2,\perp}} = X_{2,\perp} (X_{2,\perp}^T X_{2,\perp})^{-1} X_{2,\perp}^T b$$

$$\langle \Pi_{X_1} a, \Pi_{X_{2,\perp}} b \rangle = a^T \Pi_{X_1} \Pi_{X_{2,\perp}} b = a^T X_1 (X_1^T X_1)^{-1} X_1^T X_{2,\perp} (X_{2,\perp}^T X_{2,\perp})^{-1} X_{2,\perp}^T b$$

$$\text{notice } X_1^T X_{2,\perp} = X_1^T (X_2 - \Pi_{X_1} X_2) = X_1^T X_2 - X_1^T X_2 = 0$$

a and b are arbitrary vectors

Thus $C(X_1)$ and $C(X_{2,\perp})$ are orthogonal

(b) $X = [X_1, X_2] \quad \therefore C(X_1) \text{ and } C(X_2) \subset C(X)$

$$C(X_{2,\perp}) = C(X_2 - \Pi_{X_1} X_2) \subset C(X_2)$$

$$\therefore C(X_1) + C(X_{2,\perp}) \subset C(X_1) + C(X_2) \subset C(X)$$

$$\dim(X) = 1 + p = p_1 + p_2$$

$$\dim(C(X_1) + C(X_{2,\perp})) = \dim(C(X_1)) + \dim(C(X_{2,\perp})) - \dim(C(X_1) \cap C(X_{2,\perp}))$$

$$= \dim(C(X_1)) + \dim(C(X_{2,\perp})) - \dim(\{0\})$$

$$= p_1 + p_2 + 0 = 1 + p$$

$$\dim(C(X)) = \dim(C(X_1) + C(X_{2,\perp})) \text{ and } C(X_1) + C(X_{2,\perp}) \subset C(X)$$

$$\therefore C(X) = C(X_1) + C(X_{2,\perp})$$

$$* C(X_1) \perp C(X_{2,\perp}) = 0 \Leftrightarrow \langle X_1 a, X_{2,\perp} b \rangle = 0, \quad a \in \mathbb{R}^{p_1}, b \in \mathbb{R}^{p_2}$$

$$\Rightarrow a^T X_1^T X_{2,\perp} b = a^T X_1^T (I_n - \Pi_{X_1}) X_2 b$$

$$X_1^T (I_n - \Pi_{X_1}) = X_1^T - X_1^T X_1 (X_1^T X_1)^{-1} X_1^T = 0$$

$$** \text{ what is } \dim(C(X_{2,\perp}))? \quad \text{note: } X = [X_1, X_2], X_2 = [x_1'' \dots x_{p_2}'']$$

the column vectors of $X_{2,\perp}$ are linear independent.

$$0 = a_1 x_1'' + \dots + a_{p_2} x_{p_2}'' + (-a_1 \Pi_{X_1} x_1'' - \dots - a_{p_2} \Pi_{X_1} x_{p_2}'')$$

$$= a_1 x_1'' + \dots + a_{p_2} x_{p_2}'' + (b_1 x_1' + \dots + b_{p_1} x_{p_1}') = 0$$

We know that $x_1', \dots, x_{p_1}', x_1'', \dots, x_{p_2}''$ are linearly independent.

$$\therefore a_1 = \dots = a_{p_2} = 0$$

$$\therefore \dim(C(X_{2,\perp})) = p_2$$

Q2) Prove Theorem 2

The orthogonal projection Π_X onto the column space of the design matrix X is decomposed as follows:

$$\Pi_X = \Pi_{X_1} + \Pi_{X_{2,\perp}}$$

$\Pi_{X_{2,\perp}}$ is an orthogonal projection onto the column space of the matrix $X_{2,\perp}$

$$C(X) = \{ \Pi_X y : y \in \mathbb{R}^n \}$$

$$C(X_1) + C(X_{2,\perp}) = \{ \Pi_{X_1} y : y \in \mathbb{R}^n \} + \{ \Pi_{X_{2,\perp}} y : y \in \mathbb{R}^n \} \\ = \{ (\Pi_{X_1} + \Pi_{X_{2,\perp}}) y : y \in \mathbb{R}^n \}$$

$$C(X) = C(X_1) + C(X_{2,\perp})$$

$$\therefore \{ \Pi_X y : y \in \mathbb{R}^n \} = \{ (\Pi_{X_1} + \Pi_{X_{2,\perp}}) y : y \in \mathbb{R}^n \}$$

$$\therefore \Pi_X = \Pi_{X_1} + \Pi_{X_{2,\perp}} \quad \checkmark$$

Textbook Approach orthogonal

$$\text{WTS: } \Pi_{X_1} + \Pi_{X_{2,\perp}} \text{ is a } \hat{\text{projection}} \Leftrightarrow \Pi_{X_1} \cdot \Pi_{X_{2,\perp}} = 0$$

$$\Leftrightarrow C(\Pi_{X_1}) \perp C(\Pi_{X_{2,\perp}}) = 0$$

$$C(X_1) = \text{im}(\Pi_{X_1}), \quad C(X_{2,\perp}) = C(\Pi_{X_{2,\perp}}) \\ = C(\Pi_{X_1})$$

↳ refer to Theorem 1-(a)

orthogonal

what kind of $\hat{\text{projection}}$ is $\Pi_{X_1} + \Pi_{X_{2,\perp}}$?

orthogonal

$$\Pi_{X_1} + \Pi_{X_{2,\perp}} \text{ is a } \hat{\text{projection}} \text{ onto } C(\Pi_{X_1} + \Pi_{X_{2,\perp}}) = C(\Pi_{X_1}) + C(\Pi_{X_{2,\perp}}) \\ = C(X_1) + C(X_{2,\perp}) \\ = C(X)$$

orthogonal

$$\therefore \Pi_{X_1} + \Pi_{X_{2,\perp}} \text{ is a } \hat{\text{projection}} \text{ onto } C(X)$$

$$\therefore \Pi_{X_1} + \Pi_{X_{2,\perp}} = \Pi_X$$

Q3) Observe that $\text{SSR} = \|\Pi_X y\|^2 - \|\Pi_{X_1} y\|^2$. The extra sum of Square due to X_2 given X_1 is defined by:

$$R(y_2 | y_1) = \|\Pi_X y\|^2 - \|\Pi_{X_1} y\|^2$$

$$\text{Show that } R(y_2 | y_1) = y^T \Pi_{X_{2,\perp}} y$$

$$\begin{aligned} \text{(Solution)} \quad R(y_2 | y_1) &= \|\Pi_X y\|^2 - \|\Pi_{X_1} y\|^2 \\ &= \|(\Pi_{X_1} + \Pi_{X_{2,\perp}}) y\|^2 - \|\Pi_{X_1} y\|^2 \\ &= y^T \Pi_{X_1} y + y^T \Pi_{X_{2,\perp}} y - y^T \Pi_{X_1} y \\ &= y^T \Pi_{X_{2,\perp}} y = 0 \end{aligned}$$

Q4) Prove Theorem 4

$$X = (X_1, X_2) \text{ and } \beta = (\gamma_1^T, \gamma_2^T)^T$$

For partitioned regression coefficient $\gamma_2 \in \mathbb{R}^q$. In this case,

$$(a) R(\gamma_2 | \gamma_1) / \sigma^2 \sim \chi^2(q; \delta), \quad \delta = \frac{\gamma_2^T X_{2,\perp}^T X_{2,\perp} \gamma_2}{\sigma^2}$$

$$(b) SSE / \sigma^2 \sim \chi^2(n-p-1)$$

$$(c) R(\gamma_2 | \gamma_1) \text{ and } SSE \text{ are independent}$$

$$(a) R(\gamma_2 | \gamma_1) / \sigma^2 = y^T \Pi_{X_{2,\perp}} y / \sigma^2 \sim \chi^2(r; \delta) \\ \Leftrightarrow (\Pi_{X_{2,\perp}} / \sigma^2) \Sigma = (\Pi_{X_{2,\perp}} / \sigma^2) \cdot \sigma^2 I_n = \Pi_{X_{2,\perp}} : \text{idempotent}$$

$$r = \text{rank} \left(\frac{\Pi_{X_{2,\perp}}}{\sigma^2} \right) = \text{rank}(\Pi_{X_{2,\perp}}) = \text{rank}(X_{2,\perp} (X_{2,\perp}^T X_{2,\perp})^{-1} X_{2,\perp}^T) \\ = \text{trace}(X_{2,\perp}^T X_{2,\perp} (X_{2,\perp}^T X_{2,\perp})^{-1}) \\ = \text{trace}(I_q) = q$$

$$\mu^T A \mu = E[y]^T \frac{\Pi_{X_{2,\perp}}}{\sigma^2} E[y] = (X\beta)^T \frac{\Pi_{X_{2,\perp}}}{\sigma^2} (X\beta) = \gamma_2^T X_{2,\perp}^T \Pi_{X_{2,\perp}} X_{2,\perp} \gamma_2 / \sigma^2 \\ = \frac{\gamma_2^T X_{2,\perp}^T X_{2,\perp} \gamma_2}{\sigma^2}$$

$$(b) SSE / \sigma^2 \sim \chi^2(n-p-1)$$

$$y^T (I_n - \Pi_X) y / \sigma^2 = y^T A y$$

$$A \Sigma = \left(\frac{I_n - \Pi_X}{\sigma^2} \right) \cdot \sigma^2 I_n = I_n - \Pi_X : \text{idempotent}$$

$$\therefore y^T \frac{I_n - \Pi_X}{\sigma^2} y \sim \chi^2(r; \delta), \quad r = \text{rank}(A), \quad \delta = E[y]^T A E[y]$$

$$\text{rank}(A) = \text{rank} \left(\frac{I_n - \Pi_X}{\sigma^2} \right) = \text{rank}(I_n - \Pi_X) = \text{trace}(I_n - \Pi_X) \\ = \text{trace}(I_n) - \text{trace}(\Pi_X) = n - (p+1) = n - p - 1$$

$$\delta = E[y]^T A E[y] = \beta^T X^T \left(\frac{I_n - \Pi_X}{\sigma^2} \right) X \beta = 0 \quad \because (I_n - \Pi_X) X = 0$$

$$\therefore SSE / \sigma^2 \sim \chi^2(n-p-1)$$

$$(c) R(\gamma_2 | \gamma_1) = y^T \Pi_{X_{2,\perp}} y$$

$$SSE = y^T (I_n - \Pi_X) y$$

$$(I_n - \Pi_X) \Pi_X = 0, \quad C(\Pi_{X_{2,\perp}}) \subset C(\Pi_X)$$

$$\therefore (I_n - \Pi_X) \Pi_{X_{2,\perp}} = 0$$

$$\Rightarrow y^T \Pi_{X_{2,\perp}} y \text{ and } y^T (I_n - \Pi_X) y \text{ are independent}$$

Q5) Prove Theorem 5

$$H_0: \gamma_2 = 0 \quad \text{vs.} \quad H_1: \gamma_2 \neq 0$$

$$F = \frac{R(\gamma_2 | \gamma_1) / q}{SSE / (n-p-1)}$$

$$\text{under } H_0: F \sim F(q, n-p-1)$$

$$\text{under } H_1: F \sim F(q, n-p-1; \delta), \quad \delta = \frac{\gamma_2^T X_{2,\perp}^T X_{2,\perp} \gamma_2}{\sigma^2}$$

$$F = \frac{R(\gamma_2 | \gamma_1) / q}{SSE / (n-p-1)} = \frac{[R(\gamma_2 | \gamma_1) / \sigma^2] / q}{(SSE / \sigma^2) / (n-p-1)} = \frac{v_1 / q}{v_2 / (n-p-1)}$$

$$\text{under } H_0, \quad v_1 \sim \chi^2(q), \quad v_2 \sim \chi^2(n-p-1) \quad \therefore F \sim F(q, n-p-1)$$

$$\text{under } H_1, \quad v_1 \sim \chi^2(q; \delta), \quad \delta = \gamma_2^T X_{2,\perp}^T X_{2,\perp} \gamma_2 / \sigma^2, \quad v_2 \sim \chi^2(n-p-1) \\ \therefore F \sim F(q, n-p-1; \delta)$$

Q6) Prove Theorem 6

In the MLRM, we wish to test the following:

$$H_0: \gamma_2 = \gamma_2^* \quad \text{vs.} \quad H_1: \gamma_2 \neq \gamma_2^*, \quad \gamma_2^* \in \mathbb{R}^q$$

In other words, we wish to compare the full model with the Sub-model.

$$(\text{The Full Model}) \quad y = X_1 \gamma_1 + X_2 \gamma_2 + \epsilon$$

$$(\text{The Sub Model}) \quad y = X_1 \gamma_1 + X_2 \gamma_2^* + \epsilon$$

By letting $\delta_2 = \gamma_2 - \gamma_2^*$ and $y^* = y - X_2 \gamma_2^*$, the two models are re-written as

$$(\text{The Full Model}) \quad y^* = X_1 \gamma_1 + X_2 \delta_2 + \epsilon$$

$$(\text{The Sub Model}) \quad y^* = X_1 \gamma_1 + \epsilon, \quad \epsilon \sim N_n(0, \sigma^2 I_n)$$

Moreover, the hypothesis is equivalent to

$$H_0: \delta_2 = 0 \quad \text{vs.} \quad H_1: \delta_2 \neq 0$$

Show that the following is true

$$F = \frac{R^*(\delta_2 | \gamma_1) / q}{SSE^* / (n-p-1)}, \quad R^*(\delta_2 | \gamma_1) = y^{*T} \Pi_{X_{2,\perp}} y^* \\ SSE^*(\delta_2 | \gamma_1) = y^{*T} (I_n - \Pi_X) y^*$$

$$\text{under } H_0: F \sim F(q, n-p-1)$$

$$\text{under } H_1: F \sim F(q, n-p-1; \delta), \quad \delta = \frac{\delta_2^T X_{2,\perp}^T X_{2,\perp} \delta_2}{\sigma^2}$$

$$\text{under } H_0, \quad R^*(\delta_2 | \gamma_1) / \sigma^2 = y^{*T} \frac{\Pi_{X_{2,\perp}}}{\sigma^2} y^* = y^T A y$$

$$A \Sigma = (\Pi_{X_{2,\perp}} / \sigma^2) \cdot \sigma^2 I_n = \Pi_{X_{2,\perp}} : \text{idempotent}$$

$$r = \text{rank}(\Pi_{X_{2,\perp}} / \sigma^2) = \text{trace}(\Pi_{X_{2,\perp}}) = \text{trace}(I_q) = q$$

$$\delta = E[y^*]^T \Pi_{X_{2,\perp}} E[y^*] / \sigma^2 = 0$$

$$\therefore R^*(\delta_2 | \gamma_1) / \sigma^2 \sim \chi^2(q)$$

under $H_1: R^*(\delta_2 | \gamma_1) / \sigma^2 = y^* \Pi_{X_{2,\perp}} y^* / \sigma^2 \sim \chi^2(r; \delta)$
 $\because (\Pi_{X_{2,\perp}} / \sigma^2) \cdot \sigma^2 I_n = \Pi_{X_{2,\perp}} : \text{idempotent}$

$$r = \text{rank}(\Pi_{X_{2,\perp}} / \sigma^2) = \text{trace}(\Pi_{X_{2,\perp}}) = q$$

$$\delta = E[y^*]^T \Pi_{X_{2,\perp}} E[y^*] / \sigma^2 = \delta_2^T X_{2,\perp}^T X_{2,\perp} \delta_2 / \sigma^2$$

CONSTRAINED LEAST SQUARE ESTIMATION

Q7) Prove the following:

Let Π be an orthogonal projection onto the column space of X and let Π_r be the orthogonal projection onto the column space of $X(X^T X)^{-1} A^T$. Then:

$$\hat{\beta}_r = (\Pi - \Pi_r) y$$

$$\hat{\chi}_r = X(X^T X)^{-1} A^T$$

Q8) Prove that X_r is full rank

$$\begin{aligned} X_r &= X(X^T X)^{-1} A^T \\ \text{rank}(X_r) &= \text{rank}(X(X^T X)^{-1} A^T) \\ &\leq \text{rank}(A^T) = \text{rank}(A) = q \\ &= \text{rank}(X^T X(X^T X)^{-1} A^T) \\ &\leq \text{rank}(X(X^T X)^{-1} A^T) \end{aligned}$$

$\therefore \text{rank}(X_r) = q$, use the property: $\text{rank}(AB) \leq \text{rank}(A), \text{rank}(B)$

Q9) Prove Lemma 9: $\text{Ker}(Z^T) = \text{im}(Z)^\perp, Z \in \mathbb{R}^{n \times m}$

$$\begin{aligned} \text{Ker}(Z^T) &= \{x: x \in \mathbb{R}^n, Z^T x = 0\} \\ \text{im}(Z) &= \{Zy: y \in \mathbb{R}^m\} \\ \text{im}(Z)^\perp &= \{x: x \in \mathbb{R}^n, x^T Zy = 0, y \in \mathbb{R}^m\} \\ &= \{x: x \in \mathbb{R}^n, y^T Z^T x = 0, y \in \mathbb{R}^m\} \\ &= \{x: x \in \mathbb{R}^n, Z^T x = 0\} \\ &= \text{Ker}(Z^T) \end{aligned}$$

Q10) Prove Theorem 8

$A \in \mathbb{R}^{q \times (p+1)}$, A is full-rank
consider the constrained version of MLRM

$$y = X\beta_r + \varepsilon, A\beta_r = 0$$

The Constrained Least Square estimator is given by:

$$\hat{\beta}_r = \hat{\beta} - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A \hat{\beta}$$

$$\begin{aligned} \hat{\beta}_r &= \underset{\substack{\beta_r \in \mathbb{R}^{p+1} \\ A\beta_r = 0}}{\text{argmin}} \|y - X\beta_r\|^2 = \underset{\substack{\beta_r \in \mathbb{R}^{p+1} \\ A(X^T X)^{-1} X^T X \beta_r = 0 \\ (\Rightarrow X^T X \beta_r = 0)}}{\text{argmin}} \|y - X\beta_r\|^2 \end{aligned}$$

$$\begin{aligned} \text{Let } z = X\beta_r, \hat{z} = X\hat{\beta}_r &= \underset{\substack{z \in \text{Ker}(X_r^T) \\ z \in C(X)}}{\text{argmin}} \|y - z\|^2 \\ &= \underset{\substack{z \in \text{im}(X_r)^\perp \\ z \in C(X)}}{\text{argmin}} \|y - z\|^2 \\ &= \underset{z \in C(X) \cap \text{im}(X_r)^\perp}{\text{argmin}} \|y - z\|^2 = \Pi_{X_r} y - \Pi_{X_r} y \end{aligned}$$

$$\begin{aligned} X\hat{\beta}_r &= \Pi_{X_r} y - \Pi_{X_r} y \\ X\hat{\beta}_r &= X(X^T X)^{-1} X^T y - X_r (X_r^T X_r)^{-1} X_r^T y \\ \hat{\beta}_r &= (X^T X)^{-1} X^T y - (X^T X)^{-1} X^T X_r (X_r^T X_r)^{-1} X_r^T y, \quad X_r = X(X^T X)^{-1} A^T \\ &= (X^T X)^{-1} X^T y - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A (X^T X)^{-1} X^T y \\ &= \hat{\beta} - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A \hat{\beta} \end{aligned}$$

(Approach #2)

Q11) Directly prove that $\Pi - \Pi_r$ is an orthogonal projection onto $C(X) \cap \text{Ker}(X_r^T)$ - This proof is mathematically more robust

* Hint: Show that $\Pi - \Pi_r$ is a projection and that $\Pi - \Pi_r = C(X) \cap \text{Ker}(X_r^T)$

(Solution):

WTS: ① $\Pi - \Pi_r$ is symmetric and idempotent ($\Rightarrow \Pi - \Pi_r$ is a projection)
② The range space of $\Pi - \Pi_r$ is equal to $C(X) \cap \text{Ker}(X_r^T)$

(1) Check if $\Pi - \Pi_r$ is symmetric and idempotent:

$$(\Pi - \Pi_r)^T = \Pi^T - \Pi_r^T = \Pi - \Pi_r \quad \therefore \Pi - \Pi_r \text{ is symmetric}$$

$$\begin{aligned} (\Pi - \Pi_r)^2 &= \Pi^2 - \Pi \Pi_r - \Pi_r \Pi + \Pi_r^2 \\ &= \Pi - \Pi_r - \Pi_r + \Pi_r \\ &= \Pi - \Pi_r \end{aligned}$$

$\therefore \Pi - \Pi_r$ is idempotent

$\therefore \Pi - \Pi_r$ is a projection onto $C(\Pi - \Pi_r)$

(2) $C(\Pi - \Pi_r) = C(X) \cap \text{Ker}(X_r^T)$

(c) for $z \in C(\Pi - \Pi_r)$, $z = (\Pi - \Pi_r)c, c \in \mathbb{R}^n$

$$z = \{X(X^T X)^{-1} X^T - X_r (X_r^T X_r)^{-1} X_r^T\} c$$

we know that $X(X^T X)^{-1} X^T \in C(X)$ and $X_r (X_r^T X_r)^{-1} X_r^T \in C(X)$

$\therefore z \in C(X)$

$$\begin{aligned} \text{Now, consider } X_r^T z &= \{X_r^T X(X^T X)^{-1} X^T - X_r^T X_r (X_r^T X_r)^{-1} X_r^T\} c \\ X_r^T z &= A(X^T X)^{-1} X^T \cdot X_r^T X(X^T X)^{-1} X^T - A(X^T X)^{-1} X^T X_r^T X_r (X_r^T X_r)^{-1} X_r^T c = 0 \\ \therefore X_r^T z &= \{X_r^T - X_r^T\} c = 0 \\ \therefore z &\in \text{Ker}(X_r^T) \end{aligned}$$

$\therefore z \in C(X) \cap \text{Ker}(X_r^T)$

z was chosen arbitrarily. Thus, $C(\Pi - \Pi_r) \subset C(X) \cap \text{Ker}(X_r^T)$

(c) Consider an arbitrary vector $w \in C(X) \cap \text{Ker}(X_r^T)$

$$w = Xb \text{ such that } X_r^T w = X_r^T Xb = 0 \quad (\exists b \in \mathbb{R}^{p+1})$$

$$\begin{aligned} w = Xb &= X(X^T X)^{-1} X^T Xb - X_r (X_r^T X_r)^{-1} X_r^T Xb \\ &= \Pi_X Xb - \Pi_{X_r} Xb \\ &= (\Pi - \Pi_r) Xb \in C(\Pi - \Pi_r) \end{aligned}$$

$\therefore C(\Pi - \Pi_r) = C(X) \cap \text{Ker}(X_r^T)$

$\Rightarrow \Pi - \Pi_r$ is a projection onto the space $C(X) \cap \text{Ker}(X_r^T)$

(Approach #3)

Q12) Prove that $\hat{\beta}_r = \hat{\beta} - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A \hat{\beta}$ using the Lagrange multiplier method.

The problem is to find the minimizer $\hat{\beta}_r$ of

$$\|y - X\beta_r\|^T \|y - X\beta_r\| \text{ subject to } A\beta_r = 0$$

It can be solved as the following procedure, generally called the Lagrange Multiplier method.

① For any fixed $\lambda \in \mathbb{R}^q$, find the minimizer $\hat{\beta}_r(\lambda)$ of

$$\mathcal{U}(\beta_r, \lambda) = \|y - X\beta_r\|^T \|y - X\beta_r\| - \lambda^T (A\beta_r - 0)$$

② Find $\lambda^* \in \mathbb{R}^q$ satisfying $\hat{A}\beta_r(\lambda^*) = 0$. As a result, we get to know:

$$\hat{\beta}_r = \hat{\beta}_r(\lambda^*)$$

(proof)

(a) WTS: $\hat{\beta}_r = \hat{\beta}_r(\lambda^*)$

① Does $\hat{\beta}_r(\lambda^*)$ satisfy the constraint $A\beta_r = 0$

② Does it minimize the Sum of Square Error for all $\beta_r \in \mathbb{R}^{p+1}$ such that $A\beta_r = 0$? i.e.:

$$\|y - X\beta_r\|^2 \geq \|y - X\hat{\beta}_r(\lambda^*)\|^2$$

$$\begin{aligned} \|y - X\beta_r\|^2 &= \|y - X\beta_r\|^2 - (\lambda^*)^T (A\beta_r - 0) = \mathcal{U}(\beta_r, \lambda^*) \\ &\geq \mathcal{U}(\hat{\beta}_r(\lambda^*), \lambda^*) \\ &= \|y - X\hat{\beta}_r(\lambda^*)\|^2 - (\lambda^*)^T (A\hat{\beta}_r(\lambda^*) - 0) \\ &= \|y - X\hat{\beta}_r(\lambda^*)\|^2 \end{aligned}$$

$$(b) \frac{\partial \mathcal{U}}{\partial \beta_r} = 2(X^T X)\beta_r + (A^T \lambda - 2X^T y) = 0$$

$$\frac{\partial \mathcal{U}}{\partial \beta_r \partial \beta_r} = 2X^T X : \text{positive definite matrix} \\ (\because X^T X \text{ is non-singular \& non-negative definite})$$

$$\begin{aligned} \mathcal{U}(\beta_r, \lambda) &= \|y - X\beta_r\|^T \|y - X\beta_r\| - \lambda^T A\beta_r \\ &= \beta_r^T X^T X \beta_r + \beta_r^T (A^T \lambda - 2X^T y) + y^T y \end{aligned}$$

$$\frac{\partial \mathcal{U}}{\partial \beta_r} = 2X^T X \beta_r + (A^T \lambda - 2X^T y)$$

$$\begin{aligned} \frac{\partial \mathcal{U}}{\partial \beta_r} \Big|_{\beta_r = \hat{\beta}_r} &= 2X^T X \hat{\beta}_r + (A^T \lambda - 2X^T y) = 0 \\ \therefore \hat{\beta}_r(\lambda) &= -\frac{1}{2} (X^T X)^{-1} (A^T \lambda - 2X^T y) \quad \dots (*) \end{aligned}$$

From ②, we know that $\hat{A}\beta_r(\lambda^*) = 0$

$$A\hat{\beta}_r(\lambda^*) = -\frac{1}{2} A(X^T X)^{-1} A^T \lambda^* + A(X^T X)^{-1} X^T y = 0$$

$$\begin{aligned} \therefore \lambda^* &= 2(A(X^T X)^{-1} A^T)^{-1} A(X^T X)^{-1} X^T y \\ &= 2(A(X^T X)^{-1} A^T)^{-1} A \hat{\beta} \end{aligned}$$

insert $\lambda^* = 2(A(X^T X)^{-1} A^T)^{-1} A \hat{\beta}$ into (*)

$$\hat{\beta}_r(\lambda^*) = -\frac{1}{2} (X^T X)^{-1} (-2X^T y) - \frac{1}{2} (X^T X)^{-1} (A^T \cdot 2(A(X^T X)^{-1} A^T)^{-1} A \hat{\beta}) = \hat{\beta} - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A \hat{\beta}$$

Q13)

Consider $y = X\beta_r + \varepsilon$, $A\beta_r = t$

prove that the constrained LSE is given as:

$$\hat{\beta}_r = \hat{\beta} - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} (A\hat{\beta} - t)$$

Where $\hat{\beta}$ is the unconstrained LSE.

Let $A\beta_0 = t$. Then the constraint becomes: $A\beta_r = A\beta_0$

$$\Rightarrow A(\beta_r - \beta_0) = 0$$

$$\text{Let } \beta_r^* = \beta_r - \beta_0$$

$$y = X\beta_r + \varepsilon \Rightarrow y - X\beta_0 = X(\beta_r - \beta_0) + \varepsilon$$

$$\Rightarrow y^* = X\beta_r^* + \varepsilon$$

$$\begin{aligned} \hat{\beta}_r^* &= \underset{\substack{\beta_r^* \in \mathbb{R}^{p+1} \\ A\beta_r^* = 0}}{\operatorname{argmin}} \|y - X\beta_r^*\|^2 = (X^T X)^{-1} X^T y^* - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A(X^T X)^{-1} X^T y^* \\ A\beta_r^* &= 0 \end{aligned}$$

$$\begin{aligned} &= (X^T X)^{-1} X^T (y - X\beta_0) - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A(X^T X)^{-1} X^T (y - X\beta_0) \\ &= \hat{\beta} - \beta_0 - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} (A\hat{\beta} - t) \end{aligned}$$

$$\hat{\beta}_r^* = \hat{\beta}_r - \beta_0$$

$$\therefore \hat{\beta}_r = \hat{\beta} - (X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} (A\hat{\beta} - t)$$

TESTING ON GENERAL LINEAR HYPOTHESIS

Q14) Prove Theorem 12

$$(a) R(\beta | \beta_r) / \sigma^2 \sim \chi^2(q; \delta), \quad \delta = \frac{\beta_r^T A^T (A(X^T X)^{-1} A^T)^{-1} A \beta}{\sigma^2}, \quad R(\beta | \beta_r) = \| \Pi y \|^2 - \| (\Pi - \Pi_r) y \|^2$$

$$(b) \frac{SSE}{\sigma^2} \sim \chi^2(n-p-1)$$

(c) $R(\beta | \beta_r)$ and SSE are independent

(Solution)

$$(a) R(\beta | \beta_r) = y^T \Pi_r y$$

$$\therefore R(\beta | \beta_r) / \sigma^2 = y^T \left(\frac{\Pi_r}{\sigma^2} \right) y = y^T A y$$

$$A \Sigma = \frac{\Pi_r}{\sigma^2} \cdot \sigma^2 I_n = \Pi_r \text{ is idempotent}$$

$$\therefore y^T A y \sim \chi^2(r; \delta), \quad r = \text{rank}(A) \text{ and } \delta = E[y^T A E[y]]$$

$$r = \text{rank}(A) = \text{rank}(\Pi_r) = \text{trace}(X_r (X_r^T X_r)^{-1} X_r^T)$$

$$= \text{trace}(X_r^T X_r (X_r^T X_r)^{-1})$$

$$= \text{trace}(I_q) = q$$

$$X_r = X(X^T X)^{-1} A^T, \quad X_r^T X_r = A(X^T X)^{-1} A^T$$

$$\delta = E[y^T A E[y]] = \beta^T X^T \frac{\Pi_r X_r}{\sigma^2} X \beta = \beta^T X^T X_r (X_r^T X_r)^{-1} X_r^T X \beta / \sigma^2$$

$$= \beta^T A^T (A(X^T X)^{-1} A^T)^{-1} A \beta / \sigma^2$$

$$(b) SSE / \sigma^2 = y^T \left(\frac{I_n - \Pi_r}{\sigma^2} \right) y = y^T A y$$

$$A \Sigma = \left(\frac{I_n - \Pi_r}{\sigma^2} \right) \cdot \sigma^2 I_n = I_n - \Pi_r : \text{idempotent}$$

$$\therefore SSE / \sigma^2 \sim \chi^2(r; \delta), \quad r = \text{rank}(A), \quad \delta = E[y^T A E[y]]$$

$$r = \text{rank}(A) = \text{rank}(I_n - \Pi_r) = \text{trace}(I_n - \Pi_r) = \text{trace}(I_n) - \text{trace}(X(X^T X)^{-1} A^T)$$

$$= \text{trace}(I_n) - \text{trace}(X^T X (X^T X)^{-1})$$

$$= \text{trace}(I_n) - \text{trace}(I_{p+1})$$

$$= n - p - 1$$

$$\delta = E[y^T A E[y]] = \beta^T X^T \left(\frac{I_n - \Pi_r}{\sigma^2} \right) X \beta = 0 \quad \because (I_n - \Pi_r) X = 0$$

$$\therefore SSE / \sigma^2 \sim \chi^2(n-p-1)$$

(c) Let $R(\beta | \beta_r) = y^T A y$ and $SSE = y^T B y$

$$R(\beta | \beta_r) \text{ and } SSE \text{ are independent} \Leftrightarrow A \Sigma B = 0, \quad \Sigma = \text{Var}(y)$$

$$A \Sigma B = \Pi_r \cdot \sigma^2 I_n \cdot (I_n - \Pi_r)$$

$$= \sigma^2 \Pi_r (I_n - \Pi_r)$$

$$= \sigma^2 (\Pi_r - \Pi_r \Pi_r)$$

$$= \sigma^2 (\Pi_r - \Pi_r) = 0 \quad \because C(\Pi_r) \subset C(\Pi_r)$$

$\therefore R(\beta | \beta_r)$ and SSE are independent.

Q15) Prove Theorem 13

$$H_0: A\beta = 0 \text{ vs. } H_1: A\beta \neq 0$$

$$F = \frac{R(\beta | \beta_r) / q}{SSE / (n-p-1)}$$

under $H_0: F \sim F(q, n-p-1)$

under $H_1: F \sim F(q, n-p-1; \delta)$,

$$\delta = \frac{(A\beta)^T (A(X^T X)^{-1} A^T)^{-1} (A\beta)}{\sigma^2}$$

(proof)

$$F = \frac{R(\beta | \beta_r) / q}{SSE / (n-p-1)} = \frac{\frac{R(\beta | \beta_r)}{\sigma^2} / q}{\frac{SSE}{\sigma^2} / (n-p-1)} = \frac{V_1 / q}{V_2 / (n-p-1)}$$

under $H_0: A\beta = 0, \quad V_1 \sim \chi^2(q), \quad V_2 \sim \chi^2(n-p-1)$

$$\therefore F \sim F(q, n-p-1)$$

$$\text{under } H_1: A\beta \neq 0, \quad V_1 \sim \chi^2(q; \delta), \quad \delta = \frac{(A\beta)^T (A(X^T X)^{-1} A^T)^{-1} (A\beta)}{\sigma^2}$$

$$V_2 \sim \chi^2(n-p-1)$$

$$\therefore F \sim F(q, n-p-1; \delta)$$

Q16) Prove Theorem 14

In the MLRM, we wish to test the following:

$$H_0: A\beta = t \text{ vs. } H_1: A\beta \neq t$$

$$F = \frac{(A\hat{\beta} - t)^T (A(X^T X)^{-1} A^T)^{-1} (A\hat{\beta} - t) / q}{SSE / (n-p-1)}$$

under $H_0: F \sim F(q, n-p-1)$

under $H_1: F \sim F(q, n-p-1; \delta)$

$$\delta = (A\hat{\beta} - t)^T (A(X^T X)^{-1} A^T)^{-1} (A\hat{\beta} - t) / \sigma^2$$

$$A\beta = t \Rightarrow A\beta = A\beta_0 \Rightarrow A\beta - A\beta_0 = 0 \Rightarrow A\beta^* = 0, \quad \beta^* = \beta - \beta_0$$

$$F = \frac{R(\beta | \beta_r) / q}{SSE / (n-p-1)} = \frac{(y^*)^T \Pi_r (y^*) / q}{SSE / (n-p-1)}$$

$$\Pi_r = X_r (X_r^T X_r)^{-1} X_r^T = X(X^T X)^{-1} A^T (A(X^T X)^{-1} A^T)^{-1} A(X^T X)^{-1} X^T$$

$$y^* = y - X\beta_0 \quad \therefore (y^*)^T \Pi_r (y^*) = (\hat{\beta} - \beta_0)^T A^T (A(X^T X)^{-1} A^T)^{-1} A(\hat{\beta} - \beta_0)$$

$$= (A\hat{\beta} - t)^T (A(X^T X)^{-1} A^T)^{-1} (A\hat{\beta} - t)$$

$$\therefore F = \frac{R(\beta | \beta_r) / q}{SSE / (n-p-1)} = \frac{V_1 / q}{V_2 / (n-p-1)}$$

under $H_0, \quad V_1 \sim \chi^2(q), \quad V_2 \sim \chi^2(n-p-1) \quad \therefore F \sim F(q, n-p-1)$

under $H_1, \quad V_1 \sim \chi^2(q; \delta), \quad \delta = (A\hat{\beta} - t)^T (A(X^T X)^{-1} A^T)^{-1} (A\hat{\beta} - t) / \sigma^2$

$$V_2 \sim \chi^2(n-p-1)$$

$$\therefore F \sim F(q, n-p-1; \delta)$$